

# The separate universe computes a different variable: an exact criterion for $\delta N = \zeta$ in non-attractor phases

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The isotropic separate universe ( $\delta N$ ) computes  $\delta N_c$ , the e-fold count of a fluid worldline labelled by its position on the initial slice; the standard  $\delta N$  formula identifies  $\delta N_c$  with Maldacena's  $\zeta$ . We show this identification is an exact linear-order, super-Hubble change of variable,  $\delta N_c = \zeta_{L,f}[1 - I/3 + O(k_L^2/a^2 H^2)]$ , with  $I = \zeta_{L,f}^{-1} \int (\epsilon/c_s^2) \dot{\zeta}_L dt$  the  $\zeta$ -growth-weighted mean:  $\delta N_c = \zeta$  at linear order iff  $I \rightarrow 0$ . At second order the worldline label matters: the initial-position label (the separate universe's own label) composes the in-in bispectrum back to the isotropic separate-universe value,  $-5$ , for every constant  $\epsilon$ ; the final-position label differs by a translation term with zero monopole. We present one nontrivial check (matter-dominated contraction,  $I \rightarrow \epsilon$ , an  $O(1)$  identification error) plus three consistency limits – attractor slow roll and ekpyrosis ( $I = 0$ , identity map) and ultra-slow-roll ( $I = O(\sqrt{\epsilon_s \epsilon_f})$ , not computed here) – and place the criterion against the non-attractor consistency-relation and beyond- $\delta N$  gradient-expansion literature. There is no discrepancy in the physics, only in which variable  $\delta N_c$  is compared to.

## I. MOTIVATION

The squeezed limit of the primordial bispectrum is governed, in single-clock inflation, by Maldacena's consistency relation, and can equivalently be computed by the separate-universe ( $\delta N$ ) formalism, which treats the long-wavelength mode as a local shift of the background evolution [1]. In a non-attractor phase, where  $\dot{\zeta} \neq 0$  on super-Hubble scales, two independent difficulties arise. First, Namjoo, Firouzjahi & Sasaki showed that ultra-slow-roll (USR) inflation violates the consistency relation,  $f_{\text{NL}} = 5/2$ , and that  $\delta N$  built from  $N(\phi)$  alone misses this: the growing mode is an independent initial datum, requiring  $N(\phi, \pi)$  to be correctly captured [3]. This initial-data requirement was extended to  $P(X)$  non-attractors by Chen, Firouzjahi, Namjoo & Sasaki, who found  $f_{\text{NL}} = 5(1 + c_s^2)/(4c_s^2)$  and confirmed  $\delta N$  with momentum agrees [4]. Pajer, Schmidt & Zaldarriaga separately argued that in genuine attractors the squeezed bispectrum is an unobservable coordinate artifact of the long mode [5], a statement later sharpened with conformal Fermi coordinates by Dai, Pajer & Schmidt [6]. Cai *et al.* tracked  $f_{\text{NL}}$  through smooth and sharp USR transitions and found  $\delta N$  and in-in agree throughout [9]. Bravo, Mooij, Palma & Pradenas derived a generalized consistency relation for non-attractor backgrounds and showed the canonical single-field local  $f_{\text{NL}}$  is removable by a change of physical coordinates [10, 11], and Pasaglia, Hu & Motohashi confirmed  $\delta N(\phi, \pi)$ /in-in agreement for PBH-motivated USR scenarios [12]. Separately, Artigas, Grain & Vennin, and Jackson *et al.*, identified a phase-space separate-universe breakdown localized to a finite window of scales at slow-roll-to-USR transitions, traced to a momentum/initial-data mismatch [13, 14].

Second, and logically independent, a companion theory-audit note on this program found that the isotropic separate universe applied to a matter-dominated cosmological contraction ( $w = 0$ ,  $\epsilon = 3/2$ ) returns  $f_{\delta N} = -5$  (initial-position label) while the from-scratch in-in squeezed bispectrum of  $\zeta$  is  $-35/16 + (15/16)\mu^2$  [23], a monopole gap of  $25/8$  – an  $O(1)$  discrepancy, not a small correction. Cai, Xue, Brandenberger & Zhang's Eq. (37) is the correct squeezed shape function – equal to the sum of their Eqs. (27)–(32) and to Li, Quintin, Wang & Cai's Eq. (4.19) at  $c_s = 1$  [8, 20] – but their quoted amplitudes Eqs. (38)–(41) and Fig. 5, including the isocles value they print as  $-35/8$ , are uniformly twice Eq. (37) in every configuration, so the correct isocles amplitude is  $-35/16$ ; [23] locates this unprinted factor of 2 downstream of Eq. (37), in the amplitude-conversion step rather than in any vertex or commutator, and the from-scratch shape ( $-35/16 + \frac{15}{16}\mu^2$ , monopole  $-15/8$ ) then matches Cai *et al.*'s Eq. (37) monomial by monomial. Li *et al.*'s  $-35/16$  (their Eq. 5.1 at  $c_s = 1$ ) evaluates the same Eq. (4.19); their rows are an independent general- $(\epsilon, c_s)$  derivation that coincides with Cai *et al.*'s at  $\epsilon = 3/2$ , not a reuse of Cai's rows. Were the from-scratch shape uniformly twice as large instead, the monopole would be  $-15/4$  and the gap to  $f_{\delta N}^{\text{init}} = -5$  would be  $-5/4$ , a ratio of  $4/3$  rather than  $8/3$ . The linear criterion  $\lambda = 1 - \epsilon/3$  is independent of any bispectrum; only the second-order  $-5$  composition uses the in-in input. A second companion note derived the exact second-order map between the two variables and showed the gap decomposes into a linear rescaling and an explicit second-order term, both proportional to  $\epsilon$  [24]. Every prior test of  $\delta N$ /in-in agreement in the literature above is an inflationary non-attractor, where  $\epsilon \ll 1$  throughout the growth of the long mode. This note asks the question directly: is there a single criterion, applicable to both inflationary and contracting non-attractors, that predicts when the two constructions agree and by how much they

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differ when they do not.

## II. THE THREADING IDENTITY AND THE MAP

Comoving gauge, ADM variables  $N = 1 + \alpha$ ,  $N_i = \partial_i \psi + \tilde{N}_i$ . The separate universe evolves each patch as a homogeneous FRW region identified by its fluid element's position  $x_i$  on a flat, super-Hubble initial slice  $\Sigma_i$  ( $k_L \ll a_i H_i$ ); its output is the e-fold count  $\delta N_c$  of the fluid (normal) congruence, labelled by  $x_i$ , from  $\Sigma_i$  to the comoving final slice  $\Sigma_f$ . Along a fluid worldline  $\dot{x}^i = -N^i$  the fully nonlinear relation between the local expansion and the volume element gives the exact identity

$$\delta N_c(x_i) = \zeta(t_f, x_f) - \frac{1}{3} \int_{t_i}^{t_f} \partial_i N^i(t, x(t)) dt, \quad x_f = x(t_f), \quad (1)$$

with  $\delta \rho_c / \rho \propto \partial_i N^i$  up to a gradient term;  $\partial_i N^i$  is the coordinate divergence of the contravariant shift  $N^i$ , and  $u^\mu \parallel n^\mu$  (fluid four-velocity along the ADM normal) presumes a single scalar degree of freedom, i.e. no vorticity. On super-Hubble scales,  $\partial_i N^i = (\epsilon/c_s^2)\dot{\zeta} - \nabla^2 \zeta / (a^2 H)$  for the  $P(X)$  class (a genuine fluid with non-adiabatic pressure adds a term) [1, 2]. Splitting the interval at the comoving slice through  $x_i$  and integrating,

$$\delta N_c = \zeta_{L,f} \left[ 1 - \frac{I}{3} + O\left(\frac{k_L^2}{a^2 H^2}\right) \right], \quad I \equiv \frac{1}{\zeta_{L,f}} \int_{t_i}^{t_f} \frac{\epsilon}{c_s^2} \dot{\zeta}_L dt, \quad (2)$$

exact on super-Hubble scales for any single-field history from a flat, super-Hubble initial slice; the dropped gradient term is  $O(k_L^2/a_i H_i^2)$  or  $O(k_L^2/a_f H_f^2)$ , whichever end of the interval dominates.  $I \rightarrow 0$  (identity map) on any constant mode, including the attractor and the dominant ekpyrotic mode;  $1 - \lambda = I/3$  reduces to  $\epsilon/3$  for constant  $\epsilon$  (i.e.  $I \rightarrow \epsilon$ ), so  $\lambda \equiv 1 - I/3 = 1 - \epsilon/3$ , and to  $\lambda_{\text{USR}} = 1 + \epsilon_f/3 - \sqrt{\epsilon_s \epsilon_f}/3$  for USR with  $\epsilon = \epsilon_s (a/a_s)^{-6}$  from a slow-roll-frozen  $\zeta_{L,i} \neq 0$  at  $a_s$ ,  $\zeta_L \propto a^3$  thereafter. At second order, for constant  $\epsilon$  and  $c_s = 1$ , assuming  $\zeta_L(t_i) = 0$  (a pure growing mode, no constant piece on the initial slice), solving the exact ADM constraints to  $O(\zeta_L \zeta_S)$  and integrating Eq. (1) gives  $f_{\delta N} = f^{\text{in-in}}/\lambda + f_{\text{map}}$ , where  $f_{\text{map}}$  depends on how the fluid worldline is labelled. Labelled by *initial* position  $x_i$  – the label the separate universe uses, since it never learns where the fluid element ends up –

$$f_{\text{map}}^{\text{init}}(\epsilon, \mu) = \frac{5\epsilon}{4(3-\epsilon)} [(\epsilon-2) - \epsilon\mu^2], \quad (3)$$

and  $f_{\delta N}^{\text{init}} \equiv f^{\text{in-in}}/\lambda + f_{\text{map}}^{\text{init}} = -5$ , isotropic, for every constant  $\epsilon$ . Labelled instead by *final* position  $x_f = x(t_f)$ , the two labels differ by a pure translation  $\xi^i = -\int N_L^i dt$  along the long mode, adding a shear-plus-dilation term with zero monopole:

$$f_{\text{map}}^{\text{fin}}(\epsilon, \mu) = -\frac{5\epsilon}{4}(1-\mu^2) = f_{\text{map}}^{\text{init}} - \frac{5\epsilon}{4(3-\epsilon)}(1-3\mu^2), \quad (4)$$

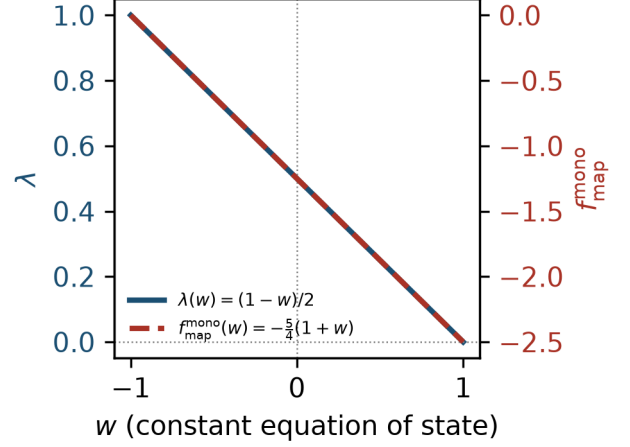


FIG. 1. The linear rescaling  $\lambda(w) = (1-w)/2$  (dimensionless, left axis) and the label-independent second-order map monopole  $f_{\text{map}}^{\text{mono}}(w) = -\frac{5}{4}(1+w)$  (dimensionless, right axis), Eq. (5), for constant equation of state  $-1 \leq w \leq 1$  at  $c_s = 1$ . At  $w = -1$  (attractor limit,  $\epsilon = 0$ )  $f_{\text{map}}^{\text{mono}}$  vanishes and  $\lambda \rightarrow 1$ , so the map is the identity;  $\lambda$  vanishes at  $w = 1$  (kination, the  $m = 3/\epsilon - 1 = 0$  constant mode), a formal limit of the constant- $\epsilon$  formula rather than a physically realized loss of linear response.

giving  $f_{\delta N}^{\text{fin}} = -\frac{15(\epsilon-4)}{4(\epsilon-3)} + \frac{15\epsilon}{4(3-\epsilon)}\mu^2$  ( $-\frac{25}{4} + \frac{15}{4}\mu^2$  at  $\epsilon = 3/2$ );  $\mu = \hat{k}_L \cdot \hat{k}_S$ . Both labels give the same, label-independent monopole  $-5\epsilon/6$ ; only the quadrupole is a labelling convention.<sup>1</sup> In terms of a constant equation of state  $w$  ( $\epsilon = \frac{3}{2}(1+w)$ ,  $c_s = 1$ ),

$$\lambda = \frac{1-w}{2}, \quad f_{\text{map}}^{\text{fin}} = -\frac{15}{8}(1+w)(1-\mu^2), \quad (5)$$

with label-independent monopole  $f_{\text{map}}^{\text{mono}} = -\frac{5}{4}(1+w)$ . Fig. 1 shows both across  $-1 \leq w \leq 1$ : at  $w = -1$  ( $\epsilon = 0$ , de Sitter/attractor limit)  $\lambda \rightarrow 1$  and  $f_{\text{map}}^{\text{mono}} \rightarrow 0$ , so the map is the identity;  $\lambda \rightarrow 0$  at  $w = 1$  (kination,  $\epsilon = 3$ , formally the  $m = 3/\epsilon - 1 = 0$  constant mode,  $\dot{\zeta}_L = 0$ ,  $I = 0 - \lambda \rightarrow 0$  is a formal limit of the constant- $\epsilon$  formula rather than a physical loss of linear response), where the general- $\epsilon$  in-in monopole  $-\frac{5}{18}(\epsilon-3)(\epsilon-6)$  itself vanishes.

## III. THE CRITERION

Equations (2)–(4) give a single statement:

*The map  $\delta N_c = \zeta_{L,f} [1 - I/3 + O(k_L^2/a^2 H^2)]$ ,  $I = \zeta_{L,f}^{-1} \int (\epsilon/c_s^2) \dot{\zeta}_L dt$ , is exact and invertible for any single-field history ( $\lambda \equiv 1 - I/3 \neq 0$ ); it is not a discrepancy in*

<sup>1</sup> Each of the five geometric contributions to  $f_{\text{map}}^{\text{fin}}$  carries an overall factor of  $\epsilon$ : the lapse, the second-order metric perturbation, the gradient term, the final-position worldline displacement, and the initial-label translation [24].

the physics, only a statement about which variable  $\delta N_c$  is. The isotropic separate universe ( $\delta N$ , with  $N(\phi, \pi)$ ) computes  $\delta N_c$ , and the standard  $\delta N$  formula identifies  $\delta N_c$  with  $\zeta$ . That identification is exact iff  $I \rightarrow 0$ ; it carries an  $O(1)$  error in the identification – not in either variable separately – whenever  $I = O(1)$ , e.g. every constant- $\epsilon$  contraction with a growing  $\zeta$ . When  $I \ll 1$  (attractor and dominant ekpyrotic mode:  $\dot{\zeta}_L = 0$ ,  $I = 0$  exactly; USR:  $\epsilon \ll 1$  throughout the growth) the identification holds to  $O(I)$ .

Via the Hamiltonian constraint on comoving slices,  $I = O(1)$  is equivalent to the long mode’s Bardeen potential growing as  $\Psi_L = O((aH/k)^2) \zeta_L$ , the textbook signature of a non-attractor contraction [17, 18], and  $I \rightarrow 0$  recovers the gradient-expansion assumption of Lyth, Malik & Sasaki [15], following Salopek & Bond [16]. The final slice used throughout is uniform- $\phi$  (comoving), which coincides with uniform- $\rho$  only at  $I \rightarrow 0$ ; at  $I = O(1)$  the two differ already at  $O(k^0)$ , so Lyth–Malik–Sasaki’s  $\delta N = \zeta_{ud}$  (uniform-density  $\rightarrow$  uniform-density) is not the object compared here, and this is a second, independent  $O(1)$  slice ambiguity of the same order as the effect this note isolates. Two logically distinct requirements are thus needed for the standard  $\delta N$  formula to give  $\zeta$  in a non-attractor phase: (i)  $N(\phi, \pi)$ , to capture the growing mode as an independent initial datum [3]; (ii)  $I \rightarrow 0$ , so that  $\delta N_c = \zeta$ . USR satisfies (ii) approximately because  $\epsilon \ll 1$  while  $\zeta$  grows – not because the comoving shift is small, which it is not: the shift itself is  $O(1/k_L)$  in every non-attractor, USR included, but its coefficient  $\epsilon \dot{\zeta}_L$  is what controls  $I$ , and this coefficient is small in USR precisely when its growing-mode initial-data problem (i) is largest. Every inflationary test cited in Sec. I checked (i) with (ii) automatically small; none tested a regime with  $I = O(1)$ .

A distinct second-order question is the choice of final e-fold surface itself: continuing the fluid-congruence field  $N_c$  onto the uniform-density surface  $\rho = \bar{\rho}(t_f)$  (rather than uniform- $\phi$ ) to second order gives  $f_{NL}^\rho = \frac{5}{24}(2\epsilon - 15)$ , isotropic, i.e.  $-\frac{5}{2}$  at  $\epsilon = \frac{3}{2}$  (initial-position label) – an  $O(1)$  slice dependence at second order as well as at first, additional to and independent of the linear-order criterion above (Appendix A 5).

Table I summarizes one nontrivial check plus three consistency limits by exact symplectic (script and manifest in the reproducibility statement). The  $w = 0$  scalar case (a  $w = 0$ ,  $c_s = 1$  field, not true dust: genuine dust has  $c_s \rightarrow 0$ , where  $I = \zeta^{-1} \int (\epsilon/c_s^2) \dot{\zeta} dt$  diverges and the criterion as stated does not apply) is the one already reported in Ref. [23]: the isotropic separate universe gives  $f_{\delta N}^{\text{init}} = -5$  for every constant  $\epsilon \in (1, 3)$ , while the in-in monopole is  $f_{\text{mono}}^{\text{in-in}} = -\frac{5}{18}(\epsilon - 3)(\epsilon - 6) = -15/8$  at  $\epsilon = 3/2$ , so the identification error,  $-5 - (-15/8) = -25/8$ , is a factor 8/3; the growing-mode monopole and the separate-universe value coincide only in the attractor limit  $\epsilon \rightarrow 0$ , where both are computed on the constant mode ( $I = 0$ ) rather than by extrapolating this  $\epsilon = 3/2$  result. USR agrees because  $\epsilon \ll 1$  throughout

the growth of the long mode – its row tests the size of the  $O(\epsilon)$  correction to the map, not a constraint from Namjoo-Firouzjahi-Sasaki’s leading-order  $f_{NL} = 5/2$  itself [3]. Attractor slow roll is the identity map, consistent with Maldacena’s consistency relation [1]. Ekpyrotic contraction ( $\epsilon \gg 3$ ) has  $I = 0$  because the physical long mode sits on the constant- $\zeta$  solution, reproducing the attractor behavior of single-field ekpyrosis found by Creminelli, Nicolis & Zaldarriaga [19]; with  $\dot{\zeta}_L = 0$  the row is a consistency check on the definition of  $I$  rather than a nontrivial test of the criterion. The non-dominant, growing- $\Psi$  ekpyrotic mode would have  $I = O(1)$  by the same criterion, but never sets  $\zeta_L(t_f)$  and so is not physically realized.  $\lambda_{\text{USR}}$  assumes  $\zeta_L$  is purely the growing  $a^3$  mode from a slow-roll-frozen value at  $a_s$ ; a slow-roll-to-USR transition with a non-negligible decaying piece is not covered.

#### IV. WHAT IS NEW

Relative to Takamizu, Mukohyama, Sasaki & Tanaka and Naruko, Takamizu & Sasaki, who computed the  $O(\epsilon_{\text{grad}}^2)$  (gradient-expansion,  $m = 2$ ) correction beyond the leading-order  $\delta N$ /separate-universe approximation for the superhorizon non-Gaussianity [21, 22], the term this note restores,  $-\nabla^2 \zeta / (a^2 H)$  in  $\partial_i N^i$ , is the same known next-to-leading-order gradient-expansion object; the point here is that in a non-attractor contraction the *other*, retained term,  $(\epsilon/c_s^2) \dot{\zeta}$ , is already  $O(k^0)$  and  $O(1)$  in  $\epsilon$ , so the leading-order ( $m = 0$ ) identification  $\delta N_c = \zeta$  can fail at  $O(1)$  without needing the  $m = 2$  correction at all; the two effects are independent and this note isolates the first one with a single closed-form criterion  $I$ . Relative to Refs. [3, 4], this note identifies a second, independent source of identification error between  $\delta N_c$  and  $\zeta$  in non-attractors – controlled by  $I$  rather than by initial data – and shows it is  $O(\epsilon)$  in USR (hence invisible to their test) and  $O(1)$  in a matter-like contraction. Relative to Ref. [5], the attractor case reduces to the vanishing of the non-attractor parameter  $\Theta \equiv \epsilon \dot{\zeta}_L / (H \zeta_L)$ , i.e.  $\Theta = 0$  and the identity map, so the two curvature variables coincide there and the only remaining question is observability; away from the attractor they already differ at  $O(1)$  before any observability argument, by an explicit, time-dependent, non-local map along the fluid worldline. Relative to Ref. [6], this note supplies the input a conformal-Fermi-coordinate treatment of a genuine non-attractor would need: the  $O(k^0)$  divergence of the long mode’s comoving shift and the resulting displacement and dilation of the local frame. Relative to Dai, Pajer & Schmidt’s “On separate universes” [7], which states the conditions under which a long-wavelength mode is locally a separate FRW universe, the assumption that fails once  $I = O(1)$  is theirs (b): the long mode is not adiabatic on the fluid worldline used here once  $\epsilon/c_s^2 \dot{\zeta}_L$  is  $O(k^0)$ , so the local patch is still a separate universe but the map from its clock to the global  $\zeta$  picks up the  $I$ -dependent rescal-

TABLE I. Validation of the criterion on four backgrounds (constant  $\epsilon$ ,  $c_s = 1$  except USR).  $I$  is the  $\zeta$ -growth-weighted mean of  $\epsilon/c_s^2$ ;  $\lambda = 1 - I/3$ ;  $f^{\text{in-in}}(\mu)$  the from-scratch in-in bispectrum and  $f_{\text{mono}}^{\text{in-in}}$  its monopole;  $f_{\delta N}^{\text{init}}$  the isotropic separate-universe value (initial-position label, the label the separate universe computes).

| case                                               | $I$     | $\lambda$                             | $f^{\text{in-in}}(\mu)$       | $f_{\text{mono}}^{\text{in-in}}$ | $f_{\delta N}^{\text{init}}$ | result                                                       |
|----------------------------------------------------|---------|---------------------------------------|-------------------------------|----------------------------------|------------------------------|--------------------------------------------------------------|
| $w=0$ scalar, $c_s=1$ , $\epsilon = 3/2$           | 3/2     | 1/2                                   | $-35/16 + \frac{15}{16}\mu^2$ | -15/8                            | -5                           | $I = O(1)$ : identification off by 25/8                      |
| USR, $\epsilon \propto a^{-6}$ (not computed here) | $\ll 1$ | $1 + O(\sqrt{\epsilon_s \epsilon_f})$ | 5/2                           | 5/2                              | 5/2                          | $I = O(\sqrt{\epsilon_s \epsilon_f})$ : agree to that order  |
| attractor slow roll                                | 0       | 1                                     | $\frac{5}{12}(1 - n_s)$       | same                             | same                         | $I = 0$ : identity map                                       |
| ekpyrosis, dominant mode                           | 0       | 1                                     | n/a ( $\dot{\zeta}_L = 0$ )   | same                             | same                         | $I = 0$ : consistency check ( $\dot{\zeta}_L = 0$ trivially) |

ing and translation of Eqs. (2)–(4) rather than reducing to the identity. Relative to Refs. [9, 12], the agreement found there is explained as the  $I \ll 1$  corner of the criterion rather than a coincidence of the USR transition. Relative to Refs. [10, 11], the coordinate-removal argument for the canonical single-field local  $f_{\text{NL}}$  is  $O(\epsilon)$ -controlled in inflation; whether an analogous physical-coordinate argument removes the  $O(1)$  in-in monopole in a matter-like contraction is a distinct, open question, not addressed here. Relative to Refs. [13, 14], the mechanism and order differ: their breakdown is localized to a finite window of scales at a slow-roll-to-USR transition and traced to a momentum mismatch, while the criterion here applies to all super-Hubble scales at  $O(k^0)$ , at constant  $\epsilon$ , with no transition required.

## V. LIMITS

The second-order map, Eqs. (3)–(5), is exact for constant  $\epsilon$  and  $c_s = 1$  (the scalar-field realization of a given  $w$ ); it is not the second-order map for a genuine fluid with  $c_s^2 = w \neq 1$ , which needs the  $c_s$ -dependent second-order ADM constraints – an open, bounded extension. The USR second-order statement in Sec. III is structural (every frozen second-order kernel is proportional to  $\epsilon$ , and vanishes as  $\epsilon \rightarrow 0$ ), not a re-solve with a time-dependent  $\epsilon(t) = \epsilon_s(a/a_s)^{-6}$ ; a full time-dependent- $\epsilon$  second-order calculation was not attempted. The linear criterion, Eq. (2), holds on super-Hubble scales for any single-field history and any  $c_s$ , from a flat, super-Hubble initial slice, up to the dropped  $O(k_L^2/a^2 H^2)$  gradient term bounded in Sec. II; it is not literally exact at all  $k_L$ .

## REPRODUCIBILITY STATEMENT

The linear criterion, second-order map, general- $w$  closed forms, and all four validations are computed by exact sympy in the committed script [research/theory\\_audit/separate\\_universe\\_](#)

[failure\\_criterion\\_2026\\_09\\_04.py](#) (SHA-256 21668ab6...), with results in the companion [.json](#) (SHA-256 4e2a49be...) and derivation in the companion note of the same base name ([.md](#)); the second-order threading map is derived in [research/theory\\_audit/threading\\_map\\_second\\_order\\_2026\\_09\\_04.{md,py,json}](#), and the in-in inputs in [research/theory\\_audit/fnl\\_monopole\\_adjudication\\_2026\\_09\\_03.md](#). The label-resolved second-order composition (both worldline labels, Eqs. (3)–(4)) and the restored gradient term (Eq. (2)) are derived and sympy-validated in [research/theory\\_audit/psu\\_gates\\_S1\\_S2\\_2026\\_09\\_04.{py,json}](#) (SHA-256 4f641022... / f4164019...). Figure 1 is generated by [make\\_fig\\_lambda\\_fmap.py](#) in this directory, transcribing the general- $w$  closed forms with no new derivation. Manifest: [reproducibility/manifests/experiments/lift2-separate-universe-failure-criterion.json](#) and [reproducibility/manifests/experiments/psu-gates-s1-s2-label-composition-criterion.json](#). Local CPU, \$0, under 5 seconds total.

## AI USAGE DISCLOSURE

The scripts verify algebra: the symbolic manipulations underlying the linear criterion, the second-order map for both worldline labels, and the four validations were performed by deterministic sympy scripts written and executed under the author’s direction, with every intermediate result printed and cross-checked against the frozen inputs. The physical identification of the separate universe’s output with the initial-position-label  $\delta N_c$  (Sec. II), and the framing of the criterion as a change of variable rather than a failure (Sec. III), are the author’s arguments, not sympy output. Claude (Anthropic) assisted with literature cross-checks, prose drafting, and LaTeX formatting under the author’s direction; all physics content, equations, and numerical results were verified by the author against the committed script output.

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## Appendix A: Self-contained second-order derivation

This appendix transcribes, from the committed companion notes cited above, the material a standalone reader needs to verify Eq. (3)–(4) and the  $-5$  composition without the external GitHub notes.

### 1. The second-order lapse/shift solution and the exact worldline identity

Comoving gauge ( $\delta\phi = 0$ ), Maldacena variables  $h_{ij} = a^2 e^{2\zeta} \delta_{ij}$ ,  $N = 1 + \alpha$ ,  $N_i = \partial_i \psi + \tilde{N}_i$ ; at first order  $\alpha = \dot{\zeta}/H$ ,  $\psi = -\zeta/H + \chi$ ,  $\partial^2 \chi = a^2 \epsilon \dot{\zeta}$ . The zero-shift threading is the fluid (normal) congruence,  $n^\mu = (1/N, -N^i/N)$ ,  $dx^i/dt = -N^i$ . The expansion of the normal congruence is  $K = \nabla_\mu n^\mu = \frac{1}{N\sqrt{h}}[\partial_t \sqrt{h} - \partial_i(\sqrt{h} N^i)]$ ; along a fluid worldline this gives the exact (fully nonlinear) identity

$$NK = \frac{d}{dt} \ln \sqrt{h} \Big|_{\text{worldline}} - \partial_i N^i. \quad (\text{A1})$$

With  $\delta N_c = \int (K/3) N dt - \tilde{N}$  and  $\ln \sqrt{h} = 3 \ln a + 3\zeta$ , this is Eq. (1) of the main text. Every difference between  $\zeta_{\text{Mald}}$  and  $\delta N_c$  is the divergence of the shift integrated along the fluid worldline: (i)  $\partial_i N^i$  itself at second order, from the second-order lapse and shift solving the exact ADM Hamiltonian and momentum constraints; (ii) its evaluation along the worldline displaced by the long mode; (iii) the choice of label (final position  $x_f$  vs. initial position  $x_i = x_f + \xi$ ), a rigid translation of the final field.

### 2. The five second-order kernel contributions (script: “threading map”)

Writing  $\delta N_c^{(2)} = \mathcal{M} \zeta_L \zeta_S$ , the kernel solving the exact ADM constraints to  $O(\zeta_L \zeta_S)$  has five pieces of distinct geometric origin, each carrying an explicit overall factor of  $\epsilon$ :

**zlap** the  $-2\zeta \partial^2 \psi$  term from  $e^{-2\zeta}$  in  $N^i = h^{ij} N_j$  (local kernel  $2\epsilon/3$ );  $f_{\text{NL}}$  contribution  $\frac{5\epsilon}{(3-\epsilon)^2}$ , monopole the same (isotropic).

**psi2** the second-order scalar shift  $\partial^2 \psi_2/a^2$  from the constraints (non-local,  $O(k_S/k_L)$  poles); contribution  $\frac{5\epsilon(-2\epsilon^2+11\epsilon-24)}{8(3-\epsilon)^2} + \frac{5\epsilon(2\epsilon^2-9\epsilon+12)}{8(3-\epsilon)^2} \mu^2$ , monopole  $\frac{5\epsilon(-\epsilon^2+6\epsilon-15)}{6(3-\epsilon)^2}$ .

**grad**  $-2\partial_i \zeta \partial_i \psi_1/a^2$ ; contribution  $-\frac{5\epsilon}{4(3-\epsilon)^2} + \frac{15\epsilon}{4(3-\epsilon)^2} \mu^2$ , monopole 0.

**wl\_fin**  $\partial_i N_S^i$  read along the worldline displaced by  $\Delta_L(t) = \int_t^{t_f} N_L dt'$ ; contribution  $\frac{5\epsilon^2}{8(3-\epsilon)^2} - \frac{15\epsilon^2}{8(3-\epsilon)^2} \mu^2$ , monopole 0.

**lab\_init** (+ **wl\_initextra**) the rigid translation  $x_f \rightarrow x_i$  (initial-position label),  $\propto (1 - 3\mu^2)$ , monopole 0.

**Totals.** Final-position label:  $f_{\text{map}} = -\frac{5\epsilon}{4} + \frac{5\epsilon}{4} \mu^2$ , monopole  $-5\epsilon/6$ . Initial-position label: monopole again  $-5\epsilon/6$ ; only the quadrupole changes, by the translation term. Every  $1/k_L$  pole cancels between the two short legs.

### 3. The two worldline labels and the translation

A worldline can be labelled by its final position  $x_f = x(t_f)$  (Eulerian, at the final comoving slice) or its initial position  $x_i = x(t_i) = x_f + \xi$ , with  $\xi^i \equiv \int_{t_i}^{t_f} N_L^i dt$  (Lagrangian, the position on the initial slice where the separate-universe patches are laid out); for the long mode  $N_L^i = a^{-2} \partial_i \chi_L$ ,  $\partial^2 \chi_L = a^2 \epsilon \dot{\zeta}_L$ , giving  $\xi(k_L) = -i \hat{k}_L \epsilon \zeta_L(k_L)/k_L$  at constant  $\epsilon$ . The two labelled fields are related by a translation by this long-mode field; in the squeezed bispectrum the  $O(k_S/k_L)$  poles cancel between the two short legs, the trace part vanishes at  $n_s = 1$ , and the shear gives the quadrupole-only translation term

$$T(\epsilon, \mu) \equiv f_{\text{map}}^{\text{init}} - f_{\text{map}}^{\text{fin}} = \frac{5\epsilon}{4(3-\epsilon)}(1-3\mu^2), \quad (\text{A2})$$

monopole 0 (all  $\epsilon$ ).

With  $\lambda = 1 - \epsilon/3$  and  $f^{\text{in-in}}(\mu, \epsilon) = \frac{5}{12}(\epsilon^2 \mu^2 - \epsilon^2 + 6\epsilon - 12)$ , the per-label maps are

$$f_{\text{map}}^{\text{fin}}(\epsilon, \mu) = -\frac{5\epsilon}{4}(1-\mu^2), \quad (\text{A3})$$

$$f_{\text{map}}^{\text{init}}(\epsilon, \mu) = \frac{5\epsilon}{4(3-\epsilon)}[(\epsilon-2) - \epsilon\mu^2], \quad (\text{A4})$$

both with monopole  $-5\epsilon/6$ , and  $f_{\delta N}^{\text{init}} = f^{\text{in-in}}/\lambda + f_{\text{map}}^{\text{init}} \equiv -5$  isotropically for every constant  $\epsilon$ . The separate universe evolves each patch as a homogeneous FRW region identified by its fluid element's coordinates on the initial slice; it has no shift and never learns where that element ends up, so its output is  $\delta N_c$  labelled by *initial* position – the label for which the composition is exactly  $-5$ .

### 4. The general- $\epsilon$ in-in shape and the identity-vs-fit check

The from-scratch in-in squeezed-limit shape used above, for general constant  $\epsilon$  [23], is

$$f^{\text{in-in}}(\mu, \epsilon) = \frac{5}{12}(\epsilon^2 \mu^2 - \epsilon^2 + 6\epsilon - 12), \quad (\text{A5})$$

whose isotropy under the composition of Sec. II depends on this full shape, not only on its monopole  $-\frac{5}{18}(\epsilon-3)(\epsilon-6)$  (the note this appendix draws on prints the monopole alone). The second-order map derivation (Secs. A1-A3 above) is computed from the exact ADM constraints alone and is validated symbolically before any in-in number is substituted: the constraint solver and the bispectrum assembly are written and frozen, then  $f^{\text{in-in}}$  is read in only at the closure step. The map derivation

therefore takes no in-in and no  $\delta N$  input; the composed monopole  $-5$  is a consequence of the map applied to the independently-derived in-in shape, not a fit to the separate universe's own  $-5$ .

### 5. The general admixture: constant-mode kernel and the corrected normalisation

For a long mode with a constant admixture,  $\zeta_L = C + G$  with  $g \equiv G_f/\zeta_{L,f}$  the growing-mode fraction and  $\lambda_g = 1 - \epsilon g/3$ , the second-order shift kernel is bilinear,  $M(g) = \frac{2}{3}\epsilon + gK_{\text{rest}}^{\text{grow}} + (1-g)K_c$ , where  $K_c$  is the constant-mode piece, obtained by re-solving the ADM constraints at  $(m_L, m_S) = (0, 3/\epsilon - 1)$ ; in the squeezed limit  $K_c \rightarrow -\frac{2}{3}\epsilon$ , independent of the short-mode history. This exactly cancels the constant-mode **zlap** contribution  $2\epsilon/3$ : a constant long mode carries no  $O(k^0)$  threading map, as it must, since a constant  $\zeta_L$  is a pure co-ordinate rescaling. The bispectrum normalisation for a mixed long mode is two-lambda, not  $1/\lambda_g^2$ : the long leg carries  $\lambda_g$ , the short (growing) leg  $\lambda_1 \equiv 1 - \epsilon/3$ ,

$$f_{\text{map}}(g) = \frac{5}{12\lambda_g\lambda_1} \frac{M(g; k_L, q)P(q) + M(g; k_L, p)P(p)}{P_S} \Big|_{\text{sq}} \\ = \frac{g\lambda_1}{\lambda_g} f_{\text{map}}(1), \quad (\text{A6})$$

so  $g = 1$  reproduces the  $-5$  (initial label) /  $-\frac{25}{4} + \frac{15}{4}\mu^2$  (final label) compositions above exactly, and  $g = 0$  gives the null map  $f_{\text{map}}(0) = 0$  for both labels – a purely constant long mode carries no super-Hubble signal (`psu_gates_S9_S10_2026.09.05.py`).

The uniform-density (-)slice question of Sec. II is distinct from this admixture: continuing  $N_c$  to second order on  $\rho = \bar{\rho}(t_f)$  gives  $f_{\text{NL}}^\rho = \frac{5}{24}(2\epsilon - 15) = -\frac{5}{2}$  at  $\epsilon = \frac{3}{2}$  (isotropic, initial label). This value does not reproduce the uniform-density separate-universe number  $-\frac{55}{16}$  quoted elsewhere in this program (gap  $\frac{15}{16}$ ). The intrinsic flat-slice initial-data term,  $f_{\text{NL}}^{\text{intr}} = \frac{5\sqrt{3}\epsilon(3-\epsilon)}{18} \frac{r_i}{W-1}$ , vanishes in the growing-mode-dominated limit that defines  $-\frac{55}{16}$ , so it does not close this gap; an independent exact separate-universe solution reproduces both  $-\frac{55}{16}$  (uniform-density slice) and  $-5$  (comoving slice) for all  $\epsilon$ , so the uniform-density  $\delta N$  value is itself a well-defined variable, not an evolution-step artifact. The gap  $5(6-\epsilon)/24$  is entirely the second-order threading map's long $\times$ short lapse monopole coefficient  $A_2 = \epsilon(3-\epsilon)^2/3$  (this map) versus the separate universe's required  $2(3-\epsilon)^2$ ; which coefficient is correct is under independent adjudication (`research/theory_audit/psu_gate_S9c_evolution_residual_2026.09.05.md`). This gap is stated as an open item, not reconciled by the slice map derived here.